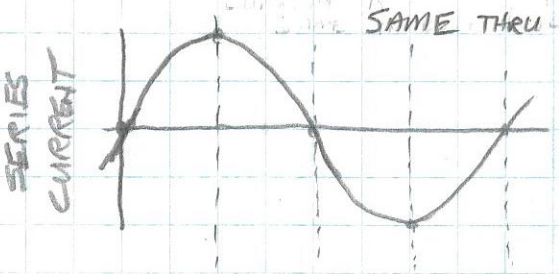
# A LOOK AT LC RESONANCE

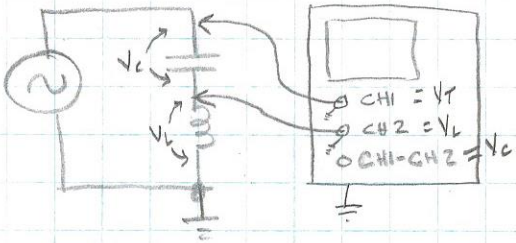
W2AEW 

## SERIES LC



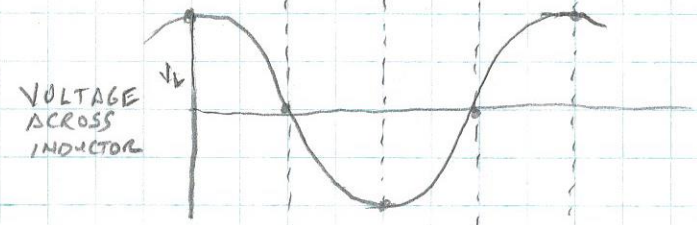

CURRENT IS SAME THRU L+C



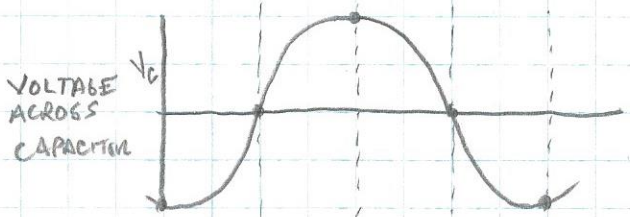




... LETS LOOK AT VOLTAGES

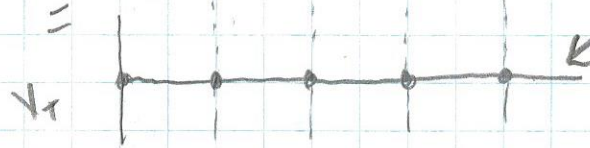
- FOR INDUCTOR, CURRENT LAGS VOLTAGE




\+ - FOR CAPACITOR, CURRENT LEADS VOLTAGE

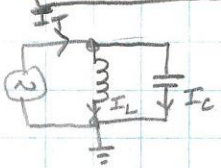



=

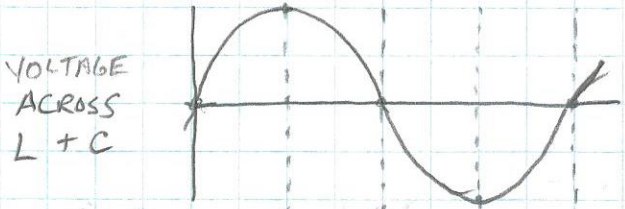


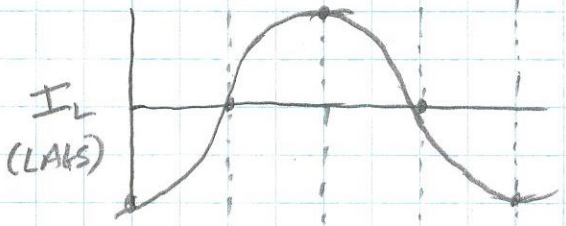

SERIES L-C LOOKS LIKE A SHORT CKT AT RESONANCE

## PARALLEL LC

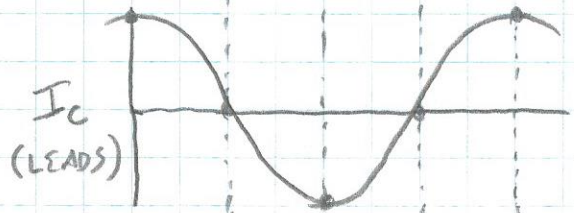



VOLTAGE IS SAME ACROSS L + C

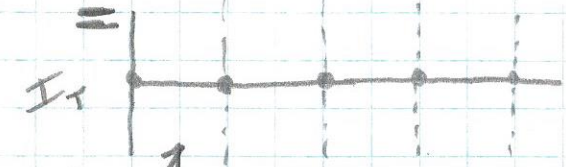



+




=




PARALLEL LC LOOKS LIKE OPEN CKT AT RESONANCE

# RESONANT CIRCUITS - PART II

VIDEO 55

W2AEW (2)




$X_L = 2\pi f L$

$X_C = \frac{1}{2\pi f C}$

RESONANCE IS WHEN $X_L = X_C$

OR..

$$2\pi f L = \frac{1}{2\pi f C}$$

$$L = \frac{1}{(2\pi)^2 f^2 C}$$

SOLVE FOR L

$$L = \frac{1}{39.48 \cdot f^2 \cdot C}$$

SOLVE FOR C

$$C = \frac{1}{39.48 \cdot f^2 \cdot L}$$

SOLVE FOR f

$$f = \frac{1}{2\pi \cdot \sqrt{LC}}$$

IN AN IDEAL WORLD, THERE ARE INFINITE COMBINATIONS OF L & C VALUES THAT WILL RESONATE AT ANY GIVEN FREQUENCY.

EXAMPLE: $f = 10\,MHz$ RESONANT FREQUENCY

START WITH 1pF CAPACITOR...

$f_R = 10\,MHz \rightarrow L = 253\,\mu H$

OR

1μF CAPACITOR, $L = 253\,pH$

1pF & 253μH

=

1μF & 253pH

- PRACTICAL, NON-IDEAL FACTORS DETERMINE WHAT TO DO...